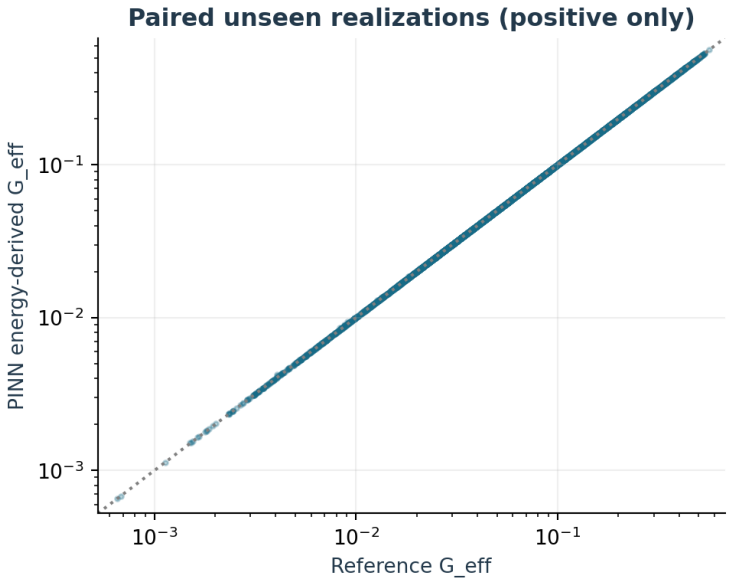
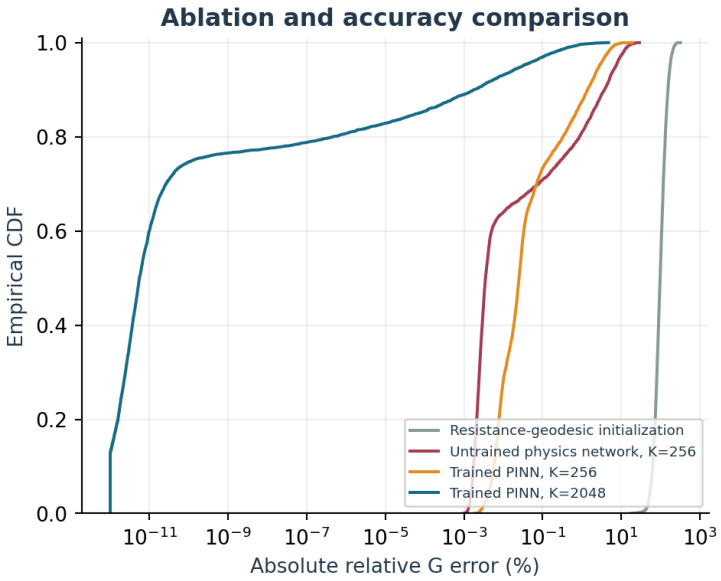
A reusable parametric graph PINN

Independent comparison with the verified random-network solver





- Main test: 9,000 unseen realizations; 4,518 carry positive flow.
- K=2048: mean relative conductance error 0.0200%.
- Weights are reused. Each input still receives fixed iterative physics updates.
- Completed study • 04 November 2026

Physical scalars do not identify a random network

The full conductance field, or a seed, must be part of the input

Input	Meaning
δ, p, σ	Distribution descriptors / neural conditioning
g field or random seed	Identifies the actual physical realization
h_left, h_right	Prescribed heads
Global conductance scale	Exact output scaling
Output	Head field, energy G estimate, flux diagnostics

- The same δ,p,σ can produce either a spanning or disconnected network.
- For a particular realization, input all 1,200 edge conductances.
- For only δ and σ, sample fields and predict ensemble statistics.
- Boundary heads and uniform conductance scaling use exact physical factors.

A discrete physics-informed architecture

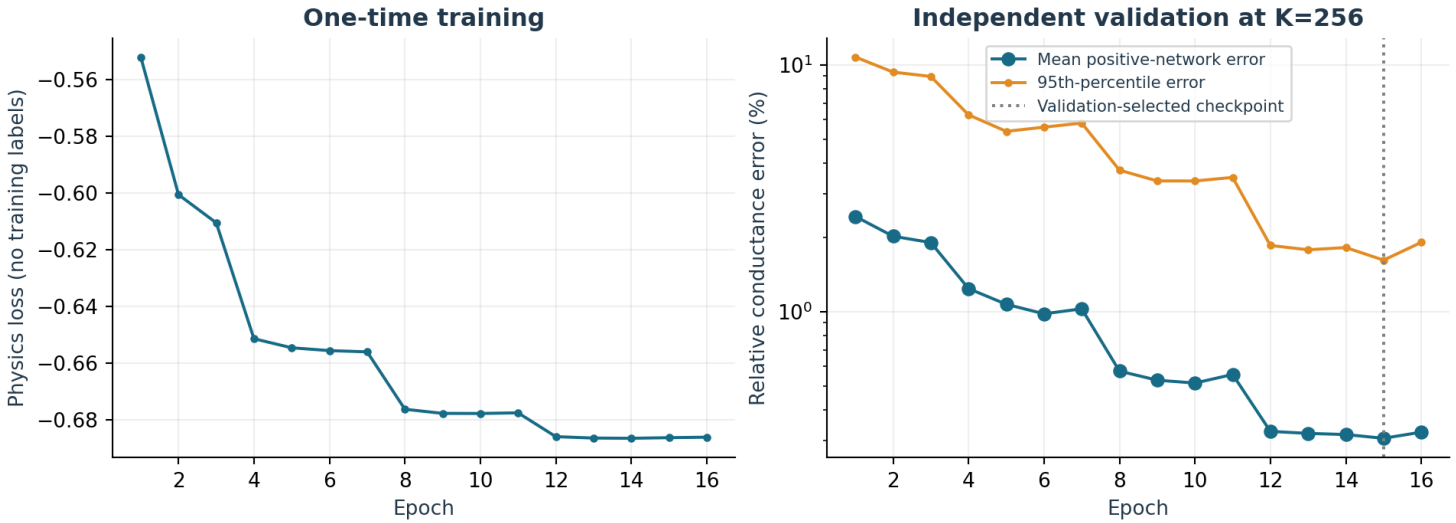
Learned encoder + parametric coefficients + fixed graph layers

Block	Definition
Node encoder	21 → 32 → 32 → 1
Coefficient MLP	6 → 16 → 2
Physics layer	h_next = h − α(h−Ph) + β(h−h_prev)
Training objective	log Dirichlet energy ratio + residual term
Trainable parameters	1,939

- Component analysis handles zero-flow and fixed-head components exactly.
- Resistance Dijkstra distances initialize head.
- Small neural networks learn initial correction and graph-level α,β.
- K residual layers update h; no new sparse solve or weight fitting.

Train once without reference solution labels

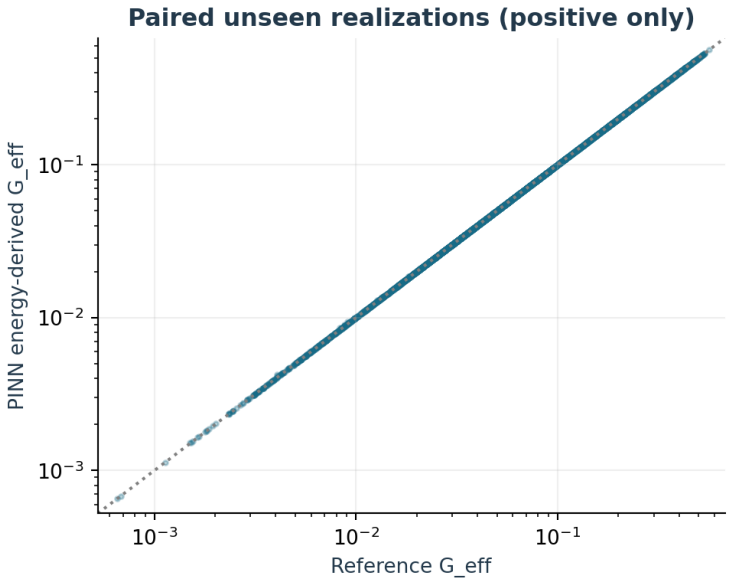
4,096 spanning training fields; 512 independent validation fields

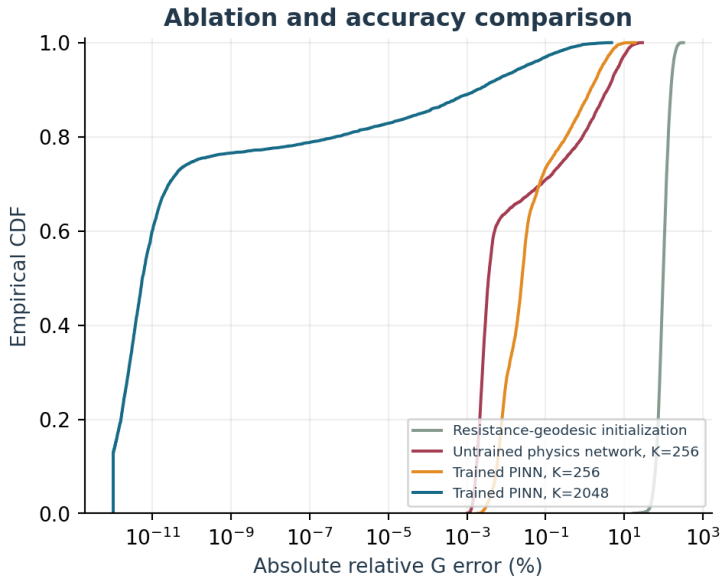


- Reference solutions are used for validation and tests only.
- 16 epochs; selected epoch 15; depth curriculum 32→64→128→256.
- Training loop 297.4s; total offline work 334.7s.
- Training, validation and test random streams are disjoint.

Accuracy improves with trained physical layers

All relative conductance errors below exclude exact zero-flow cases

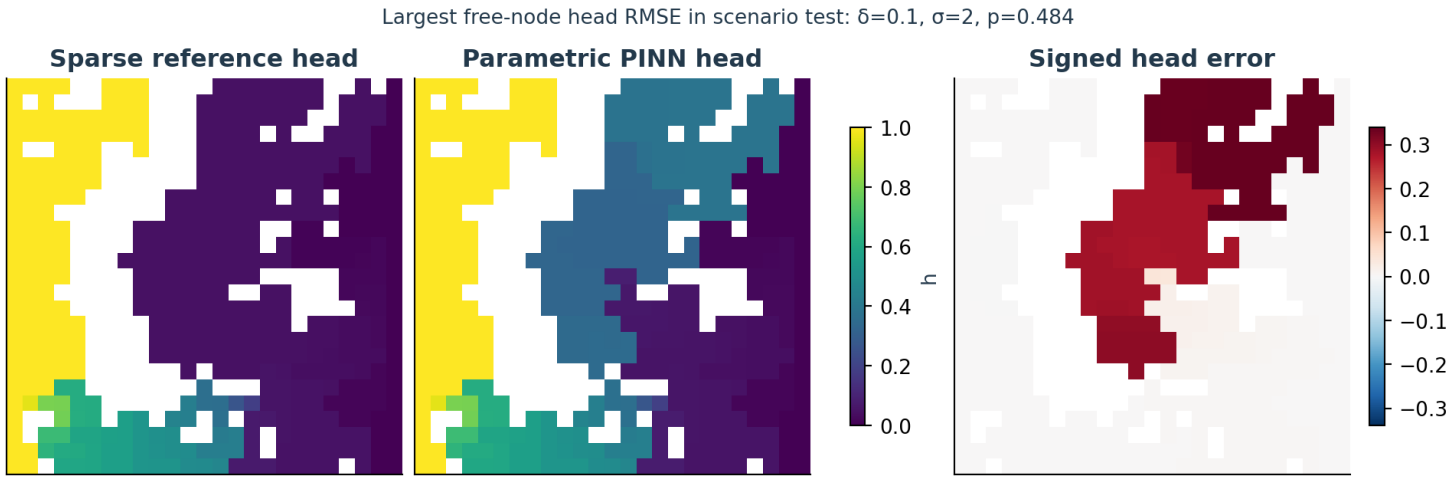




- K=256 trained mean error: 0.4705%.
- Untrained K=256 mean error: 1.1381%.
- K=2048: mean 0.0200%, p95 0.0393%.
- Maximum K=2048 error: 4.739%; accuracy is not uniform.

Energy accuracy and field accuracy differ

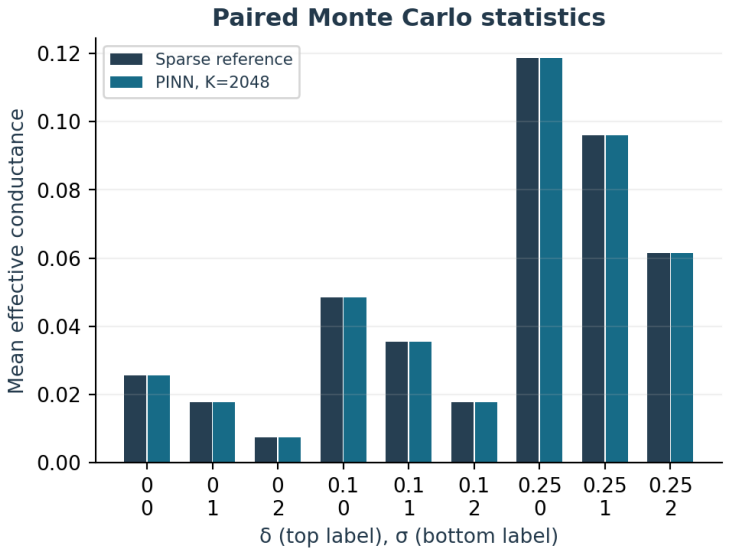
A deliberately difficult held-out head-field example

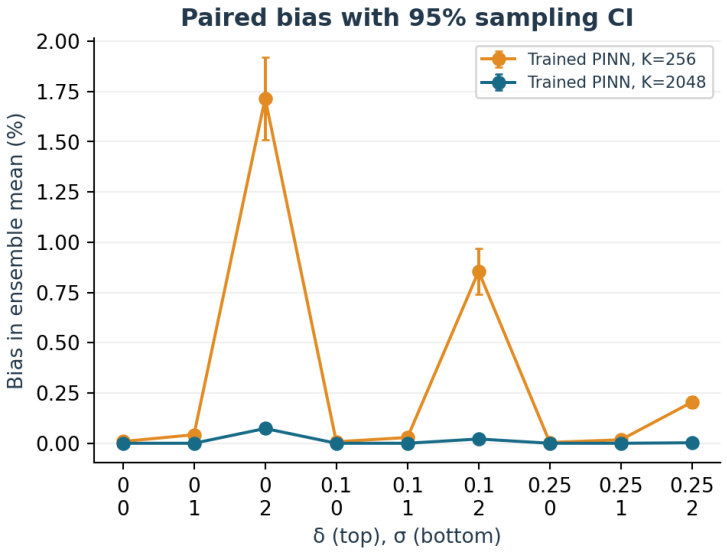


- G is estimated from Dirichlet energy, an admissible-head upper bound.
- Approximate inlet flux can differ from the energy estimate.
- Mean free-node head RMSE: 0.001460.
- 95th-percentile inlet/outlet imbalance: 3.02%.

Paired uncertainty comparison

Same physical fields for sparse reference and neural prediction

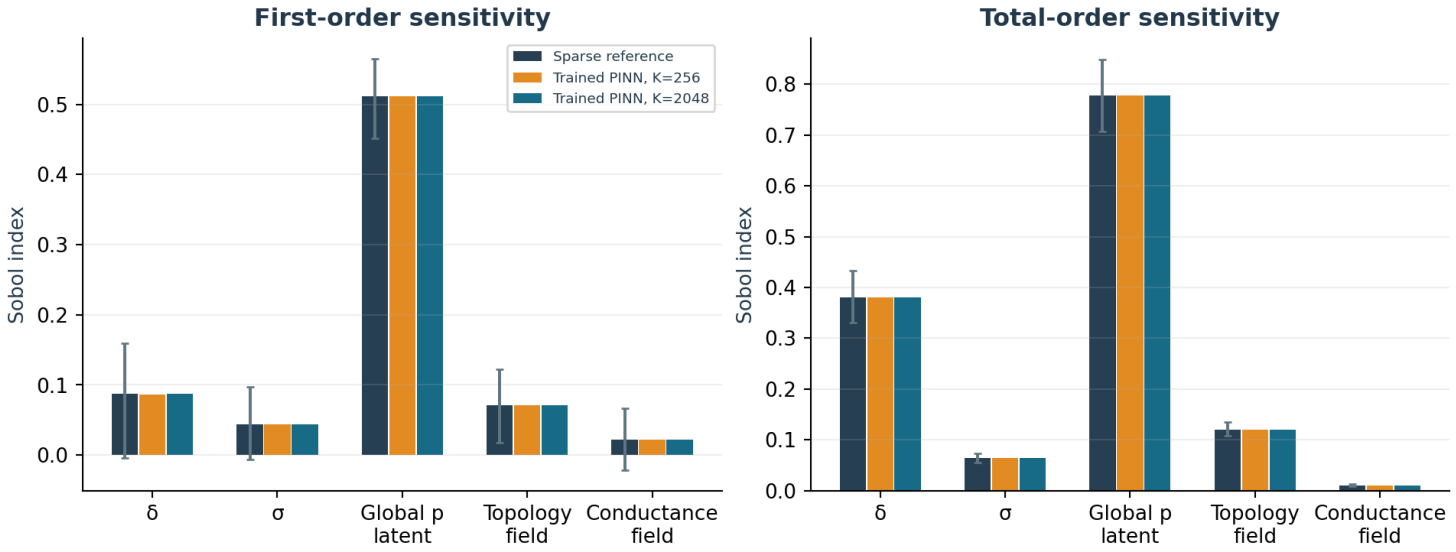




- Nine scenarios × 1,000 new realizations each.
- Compare mean, standard deviation, median, upper quantile and conditional flow.
- P(G=0) agrees exactly through structural connectivity preprocessing.
- Old 10,000-sample study and new reference means differ by Monte Carlo variation.

Sensitivity rankings are reproduced

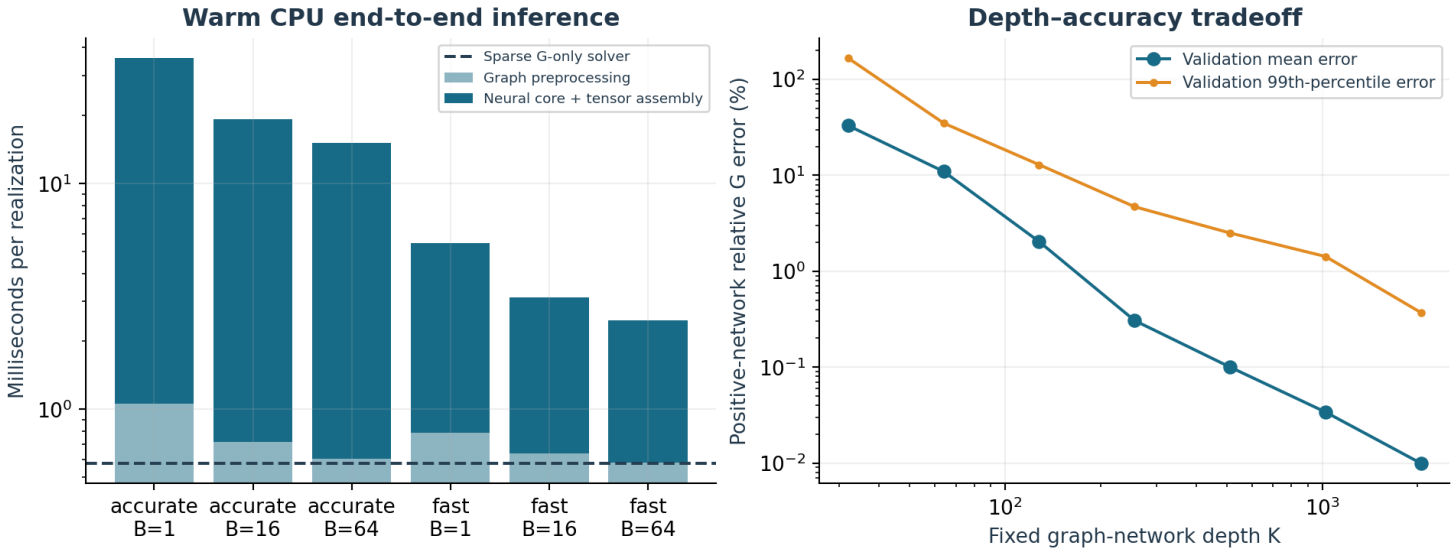
2,048 independent base pairs; 14,336 matched physical fields



- Five groups: δ, σ, global-p latent U, topology W and conductance Z.
- 1,000 paired bootstrap resamples preserve input coupling.
- Compare model-minus-reference indices on the same fields.
- Total indices overlap; their sums are not independent variance shares.

Reusable weights do not guarantee a speedup

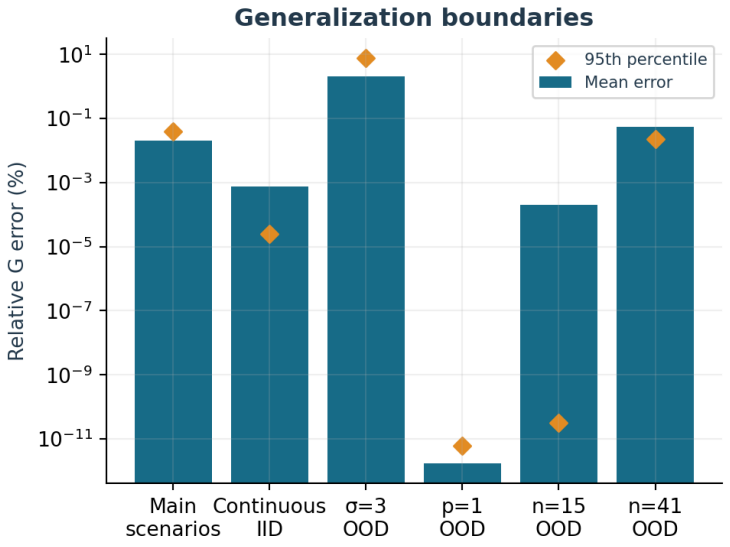
Measured warm CPU cost includes graph preprocessing and tensor assembly

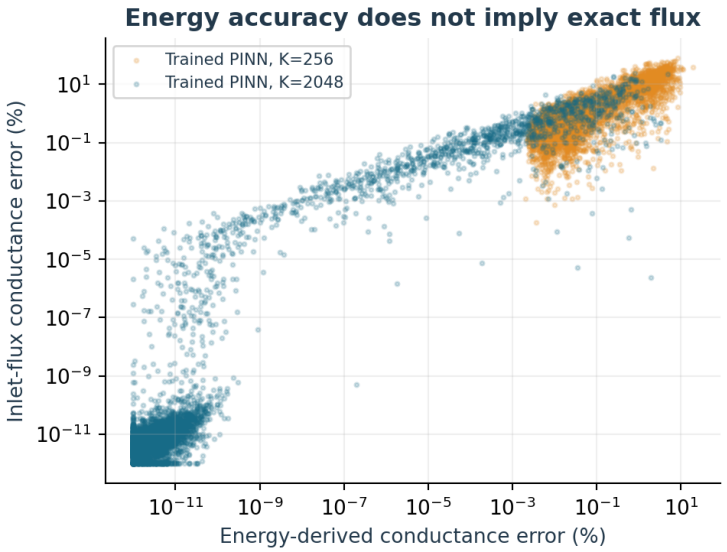


- Sparse G-only solver: 0.576 ms per realization.
- Fast PINN, batch 64: 2.508 ms per realization.
- Accurate PINN, batch 64: 15.097 ms per realization.
- For this 25×25 problem, direct solving is faster; no positive break-even was observed.

Generalization has boundaries

Stress tests do not imply a guarantee outside the training domain





- At σ=3, mean relative G error rises to 2.08%.
- The σ=3 maximum error is 20.26%.
- Other tests: p=1 and n=15,41; same learned weights, no refitting.
- Strong contrast and weak anchoring can require much larger depth.

Use the trained model directly

The default prediction path does not call the sparse solver

Task	Command from repository root
One realization	python parametric_pinn/scripts/predict.py --p .6 --sigma 1
Ensemble	.../predict.py --delta .25 --sigma 1 --draws 1000
Supplied field	.../predict.py --conductances field.npy
Optional reference check	.../predict.py --compare-reference
Complete reproduction	make -C parametric_pinn all

- Select fast K=256 or accurate K=2048.
- Provide a .npy conductance field, or distribution parameters plus a seed.
- Use multiple draws to obtain stochastic summaries.
- Keep inlet flux diagnostics separate from the energy-derived G estimate.

Conclusions and separate deliverables

36 automated tests; reproducible fields, checkpoints and paired comparisons

Artifact	Location
Reports	reports/comparison_fa.pdf; comparison_en.pdf
Trained weights	models/parametric_pinn.pt
Paired results and timings	results/
Executed notebook	notebooks/comparison.ipynb
API and reproduction	README.md; scripts/

- The graph PINN reuses trained parameters across unseen random fields.
- High in-domain conductance accuracy does not imply exact local conservation.
- The direct solver retains the advantage in precision and measured CPU cost here.
- All new outputs are isolated under parametric_pinn/.